Li JING

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Education

Massachusetts Institute of Technology Cambridge, MA
Ph.D. in Physics Sept. 2014 - Dec. 2019
Advisor: *Marin Soljacic*
Thesis: *Physical Symmetry Enhanced Neural Networks*

Peking University Beijing, China
B.S. in Physics Sept. 2010 - Dec. 2014
B.A. in Economics (double degree) Sept. 2012 - Dec. 2014

Experiences

Facebook AI Research New York, NY
Postdoctoral Researcher Feb. 2020 - now
Advised by Yann LeCun
- developed state-of-the-art self-supervised learning methods for visual representation learning
- invented regularization method for representation learning based on implicit regularization
- investigated semi-supervised learning, multimodal learning
- theoretically studied contrastive self-supervised learning dynamics

Massachusetts Institute of Technology Cambridge, MA
Research Assistant Sept. 2016 - Dec. 2019
Advised by Marin Soljacic
- invented unitary recurrent neural networks with long-term memory ability
- applied hybrid unitary recurrent neural networks to NLP applications
- investigated equivariant neural network architectures
- pioneered the direction AI methods for optical material design

Lightelligence Inc. Boston, MA
Co-founder, Algorithm Team Lead Jul. 2018 -Jan. 2019
AI-hardware startup, B-series
- invented quantization algorithm for low-precision noisy neural networks
- led a team building optical computing unit SDK

Massachusetts Institute of Technology Cambridge, MA
Research Assistant Sept. 2014 - Jul. 2016
Advised by Wolfgang Ketterle
- conducted research in large-scale experimental quantum physics

Peking University Beijing, China
Undergrad Research Assistant Jan. 2011 -Jun. 2014
Advised by Heng Fan
- conducted research in theoretical quantum information and quantum cryptography

Selected Honors and Awards

Forbes China 30 under 30 - *Enterprise Technology* 2019
International Physics Olympiad (**IPhO**) - *Gold Medal* Zagreb, Croatia, 2010

Publications

1. Jure Zbontar*, **Li Jing***, Ishan Misra, Yann LeCun, Stéphane Deny, “Barlow Twins: Self-Supervised Learning via Redundancy Reduction”, International Conference on Machine Learning (**ICML**) (2021)

2. Zhedong Wang, Chao Qian, Tong Cai, Longwei Tian, Zhixiang Fan, Jian Liu, Yichen Shen, Li Jing, Jianming Jin, Er-Ping Li, Bin Zheng, Hongsheng Chen, “Demonstration of Spider‐Eyes‐Like Intelligent Antennas for Dynamically Perceiving Incoming Waves”, Advanced Intelligent Systems (2021)

3. Rumen Dangovski, Michelle Shen, Dawson Byrd, **Li Jing**, Desislava Tsvetkova, Preslav Nakov, Marin Soljacic, “We Can Explain Your Research in Layman's Terms: Towards Automating Science Journalism at Scale”, AAAI Conference on Artificial Intelligence (**AAAI**) (2021)

4. **Li Jing**, Jure Zbontar, Yann LeCun, “Implicit Rank-Minimizing Autoencoder”, Conference on Neural Information Processing Systems (**NeurIPS**) (2020)

5. Matthew Khoury, Rumen Dangovski, Longwu Ou, Yichen Shen, Preslav Nakov, **Li Jing**, “Vector-Vector-Matrix Architecture: A Novel Hardware-Aware Framework for Low-Latency Inference in NLP Applications”, Conference on Empirical Methods in Natural Language Processing (**EMNLP**) (2020)

6. Samuel Kim, Peter Lu, Srijon Mukherjee, Michael Gilbert, **Li Jing**, Vladimir Ceperic, Marin Soljacic, “Integration of Neural Network-Based Symbolic Regression in Deep Learning for Scientific Discovery” IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**) (2020)

7. Thomas Christensen, Charlotte Loh, Stjepan Picek, Domagoj Jakobovic, **Li Jing**, Sophie Fisher, Vladimir Ceperic, John Joannopoulos, Marin Soljacic, “Predictive and generative machine learning models for photonic crystals”, **Nanophotonics** (2020)

8. Mihika Prabhu, Charles Roques-Carmes, Yichen Shen, Nicholas Harris, **Li Jing**, Jacques Carolan, Ryan Hamerly, Tom Baehr-Jones, Michael Hochberg, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljacic, “A Recurrent Ising Machine in a Photonic Integrated Circuit”, **Optica**, (2020)

9. Chao Qian, Bin Zheng, Yichen Shen, **Li Jing**, Erping Li, Lian Shen, Hongsheng Chen, “Deep learning enabled self-adaptive invisibility cloak”, **Nature Photonics** (2020)

10. Charles Roques-Carmes, Yichen Shen, Cristian Zanoci, Mihika Prabhu, Fadi Atieh, **Li Jing**, Tena Dubcek, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljacic, “Heuristic Recurrent Algorithms for Photonic Ising Machines”, **Nature Communications** (2020)

11. Rumen Dangovski*, **Li Jing***, Marin Soljacic, “Rotational Unit of Memory: A Novel Representation Unit for RNNs with Scalable Applications”, Transactions of the Association for Computational Linguistics (**TACL**) (2019)

12. Yurui Qu*, **Li Jing***, Yichen Shen, Min Qiu, Marin Soljacic, “Migrating Knowledge between Physical Scenarios based on Artificial Neural Networks”, **ACS Photonics** (2019)

13. **Li Jing***, Caglar Gulcehre*, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljacic, Yoshua Bengio, "Gated Orthogonal Recurrent Units: On Learning to Forget", **Neural Computation** (2019)

14. John Peurifoy, Yichen Shen, **Li Jing**, Yi Yang, Fidel Cano-Renteria, Brendan Delacy, John Joannopoulos, Max Tegmark, Marin Soljacic, "Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks", **Science Advances** (2018)

15. **Li Jing***, Yichen Shen*, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljacic, "Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs", Proceedings of International Conference on Machine Learning (**ICML**) (2017)

16. Heng Fan, Yi-Nan Wang, **Li Jing**, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu, "Quantum cloning machines and the applications", **Physics Reports** (2014)

17. Yong-Liang Zhang, Huan Wang, **Li Jing**, Liang-Zhu Mu, Heng Fan, "Fitting magnetic field gradient with Heisenberg-scaling accuracy", **Scientific Reports** (2014)

18. Yong-Liang Zhang, Yi-Nan Wang, Xiang-Ru Xiao, **Li Jing**, Liang-Zhu Mu, VE Korepin, Heng Fan, "Quantum network teleportation for quantum information distribution and concentration", **Physical Review A** (2013)

19. **Li Jing**, Yi-Nan Wang, Han-Duo Shi, Liang-Zhu Mu, Heng Fan, "Minimal input sets determining phase-covariant and universal quantum cloning", **Physical Review A** (2012)

20. Zhao-Xi Xiong, Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Jin Lei, Liang-Zhu Mu, Heng Fan, "General quantum key distribution in higher dimension", **Physical Review A** (2012)

21. Yi-Nan Wang, Han-Duo Shi, **Li Jing**, Zhao-Xi Xiong, Jin Lei, Liang-Zhu Mu, Heng Fan, "Non-compression of quantum phase information", **Journal of Physics A: Mathematical and Theoretical** (2012)

22. Yi-Nan Wang, Han-Duo Shi, Zhao-Xi Xiong, **Li Jing**, Xi-Jun Ren, Liang-Zhu Mu, Heng Fan, "Unified universal quantum cloning machine and fidelities", **Physical Review A** (2011)

Pre-prints

1. Ileana Rugina, Rumen Dangovski, Renbin Liu, **Li Jing**, Preslav Nakov, Marin Soljacic, "Data-Informed Global Sparseness in Attention Mechanisms for Deep Neural Networks", arXiv: 2012.02030

2. Evan Vogelbaum, Rumen Dangovski, **Li Jing**, Marin Soljačić, "Contextualizing Enhances Gradient Based Meta Learning", arXiv: 2007.10143

3. **Li Jing***, Rumen Dangovski*, Marin Soljacic, "WaveletNet: Logarithmic Scale Efficient Convolutional Neural Networks for Edge Devices", arXiv:1811.11644

4. Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Ru-Quan Wang, Liang-Zhu Mu, Heng Fan, "Quantum key distribution based on a quantum retrodiction protocol", arXiv: 1205.3556

Patents

1. Charles Roques-armes, Yichen Shen, Li Jing, Tena Dubcek, Scott A Skirlo, Hengameh Bagherianlemraski, Marin Soljacic, "Optical Ising machines and optical convolutional neural networks", US patent number 11,017,309, issued in May 2021

Invited Talks

1. Barlow Twins: Self-supervised Learning via Redundancy Reduction
   International Conference on Machine Learning (ICML) Virtual, Jul. 2021

2. Implicit Rank-Minimizing Autoencoder
   Conference on Neural Information Processing Systems (NeurIPS) Virtual, Dec. 2020

3. Physics Inspired Deep Learning Algorithms
   Facebook AI Research New York, NY, Nov. 2019

4. Phase-encoded Recurrent Neural Networks and their Applications
   DeepMind Mountain View, CA, Oct. 2019

5. Phase-encoded RNNs and Scientific Text Summarization
   Amazon Alexa AI Cambridge, MA, Sept. 2019

6. Photonics and AI: Algorithms and Applications
   APS Physics of AI conference Yorktown Heights, NY, Apr. 2019

7. **MIT-CHIEF Annual Conference, AI panel** Cambridge, MA, Nov. 2018

8. Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks
   Conference on Neural Information Processing Systems (NIPS) - Deep Learning for Physical Science Workshop Long Beach, CA, Dec. 2018

9. Gated Orthogonal Recurrent Units: On Learning to Forget
   AAAI Conference on Artificial Intelligence (AAAI) - Workshop on Reasoning and Learning for Human-Machine Dialogues New Orleans, LA, Jan. 2018

10. Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs
    International Conference on Machine Learning (ICML) Sydney, Australia, Jul. 2017

Press Coverage

1. **Techacute**: Self-Adaptive Invisibility Cloak: The "Magic" in Optics 2020

2. **SciTechDaily**: Solving complex problems at the speed of light 2020

3. **MIT News**: Can science writing be automated? A neural network can read scientific papers and render a plain-English summary. 2019

4. **MIT News**: AI-based method could speed development of specialized nanoparticles. Neural network could expedite complex physics simulations. 2018

Professional Services

Conference reviewer:
International Conference on Learning Representations (ICLR)
International Conference on Machine Learning (ICML)
Conference on Neural Information Processing Systems (NeurIPS)
Conference on Empirical Methods in Natural Language Processing (EMNLP)

Journal reviewer:
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
IEEE Computational Intelligence Magazine (CIM)
IEEE Photonic Technology Letters
Proceedings of the National Academy of Sciences of the United States of America (PNAS)
Physical Review Letters
Physical Review A
Physical Review X
Physical Review Research
Physical Review Applied
Scientific Reports
Journal of Applied Physics
Optics Express
Optica
Advanced Photonics
Nanophotonics
ACS Photonics
Nature Photonics

Mentoring

1. Pawan Goyal (undergrad at MIT): Efficient architecture for video understanding 2019

2. Lay Jain (undergrad at MIT, now quantitative researcher at Citadel Securities): Equivariance in CNN 2019

3. Alvaro Inesta (Exchange master student at MIT): Pretraining for differential privacy 2018

4. Ivan Ivanov (RSI program, now undergrad at Cambridge University): Efficient unitary recurrent neural networks 2017

5. Rawn Henry (undergrad at MIT, now AI Engineer at Nvidia): Nonlinearity for spectral pooling 2017

6. John Peurifoy (undergrad at MIT, now CEO at Floating point group): Unitary RNNs, AI application for physics 2017

7. Rumen Dangovski (undergrad at MIT, now PhD student at MIT EECS): Orthogonal RNNs with associative memory, NLP applications 2017